NSBM GREEN UNIVERSITY

Faculty of Computing

Web and Mobile Application Development — Batch 26.1

Green Campus

Smart Campus Beautification and Community Engagement System

FINAL PROJECT REPORT

GROUP U

Submitted in partial fulfilment of the requirements of the Web and Mobile Application Development module

Field	Details
Project Title	Green Campus – Smart Campus Beautification and Community Engagement System
Team Leader	K R A S Meedum
Module	Web and Mobile Application Development
Lecturer	Ms. Thisarani Wickramasinghe
Batch	26.1
Submission Date	2nd September 2026

Group Members

#	Full Name	Student ID	Role / Contribution
1	K R A S Meedum	39967	HTML, CSS, JavaScript + Flyer + Report + Presentation + Web Hosting & Deployment
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3	Bandaranayake G N	40585	PHP + Flyer + Presentation
4	A U Y Pathirana	37611	HTML, CSS, JavaScript + Report + Presentation
5	U H L Colambage	37557	MySQL + Report + Presentation
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1. Introduction

Modern university campuses generate large amounts of waste and face ongoing challenges in maintaining clean, green, and sustainable environments. At the same time, students are often willing to contribute to campus improvement initiatives but lack a centralized platform through which they can submit ideas, discover volunteering opportunities, and track their own participation. Green Campus was developed to address this gap as our submission for Project Topic 3 of the Web and Mobile Application Development final project: "Green Campus – Smart Campus Beautification and Community Engagement System."

Green Campus is a full-stack web application built using HTML, CSS, JavaScript, PHP, and MySQL. It provides two distinct interfaces — an Admin Panel for university staff who manage projects, campaigns and student submissions, and a Student Panel through which students register, submit improvement ideas, and join environmental volunteering campaigns. The system implements complete CRUD (Create, Read, Update, Delete) functionality, secure authentication, and relational database integration, in line with the requirements set out in the module's final project assignment.

2. Project Objectives

- To provide a centralized digital platform for managing campus beautification and sustainability initiatives.
- To allow students to submit campus improvement ideas and track their approval status.
- To allow students to discover and register for volunteering campaigns with real-time capacity tracking.
- To give administrators full control over projects, campaigns, and idea moderation through a dedicated dashboard.
- To demonstrate secure, role-based authentication separating Admin and Student access.
- To apply complete CRUD operations backed by a normalized MySQL database.
- To produce a responsive, user-friendly interface consistent with modern web design practices.

3. System Description

3.1 Overview

Green Campus is organized around two roles — Administrator and Student — each with a dedicated dashboard. After registering or logging in, users are redirected based on their role: administrators reach a single-page dashboard with sidebar navigation between Dashboard, Ideas, Projects, Campaigns, Students, Registrations and Reports sections, while students are directed to a dashboard that surfaces their personal statistics, submitted ideas, ongoing projects and upcoming campaigns.

3.2 User Roles and Features

Role	Key Features
Administrator	Login; create/delete campus improvement projects (add_project.php, admin_dashboard.php); create/delete volunteering campaigns with a maximum volunteer limit (add_campaign.php); approve or reject student-submitted ideas (update_idea.php); view registered students and campaign registrations; view auto-generated participation/impact reports.
Student / User	Register and login (register.php, login.php); submit campus improvement ideas (submit_idea.php); browse and join volunteering campaigns with live seat availability (campaigns.php, dashboard.php); view all campus projects and their progress (projects.php); track personal participation history (ideas submitted, campaigns joined) from the dashboard.

4. Technology Stack

Layer	Technology
Frontend	HTML CSS JavaScript using vs code and notebook
Backend	PHP notebook
Database	MySQL (phpMyAdmin)
Server Environment	XAMPP

5. Database Design

The system uses a relational MySQL database (green campus) consisting of six core tables. The users table stores login credentials and role information for both administrators and students, while the students' table stores extended student-specific records used by the volunteering and idea submission modules.

Table	Purpose	Key Fields
users	Authentication and role management	ID, name, email, password (hashed), role (admin/student)
students	Student profile records linked to users	ID, name, email, password, created_at
projects	Campus improvement projects created by admins	ID, title, description, category, progress, status, created_at
campaigns	Volunteering campaigns created by admins	ID, title, description, location, campaign_date, max_volunteers, created_at
registrations	Join table linking students to campaigns	ID, student ID, campaign, ID registered ID
ideas	Student-submitted improvement ideas	ID, student ID, title, description, category, status (pending/approved/rejected), created_at

Relationships: registrations links students to campaigns in a many-to-many fashion (one student can join many campaigns, one campaign can have many student registrations), while ideas have a one-to-many relationship from students to their submitted ideas. Foreign key constraints ensure referential integrity, and cascading deletes are handled at the application level — for example, deleting a campaign from the admin dashboard also removes its associated registrations.

6. Key Features Implemented

6.1 Authentication and Access Control

- Student self-registration (register.php) that creates linked records in both the users and students tables, with duplicate-email checks and password hashing.
- Unified login (login.php) that authenticates against the users table and redirects to the Admin or Student dashboard based on role.
- Route protection on every page via session checks, redirecting unauthenticated visitors to login.php.
- Secure logout (logout.php) that destroys the session.

6.2 Idea Submission and Moderation (CRUD)

- Students submit improvement ideas with a title, description and category (submit_idea.php); new ideas are stored with a status of 'pending'.
- Administrators review all submitted ideas and approve or reject them with a single click (update_idea.php), updating the idea's status in the database.

6.3 Project Management (CRUD)

- Administrators create new campus improvement projects with a category, progress percentage and status (add_project.php).
- All users can browse the full list of projects with a live progress bar and status badge (projects.php).
- Administrators can delete projects directly from the dashboard (admin_dashboard.php).

6.4 Volunteering Campaigns and Registration (CRUD)

- Administrators create campaigns with a title, description, location, date and maximum volunteer capacity (add_campaign.php).
- Students browse open campaigns and join with one click; the system checks for duplicate registrations and enforces the maximum volunteer limit before inserting a new registration (campaigns.php, dashboard.php).
- Already-registered students see a confirmation badge, and campaigns that have reached capacity are automatically disabled ('Campaign Full').
- Administrators can delete a campaign, which also removes all of its associated registrations (admin_dashboard.php).

6.5 Admin Dashboard and Reporting

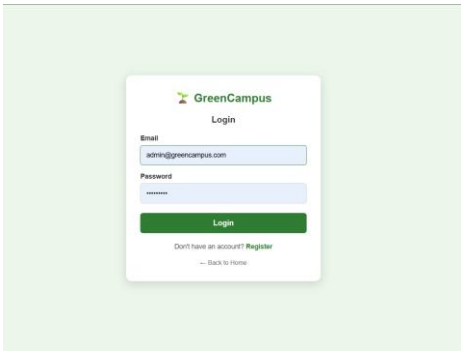
- A single sidebar-driven dashboard (admin_dashboard.php) with sections for Dashboard, Ideas, Projects, Campaigns, Students, Registrations and Reports.

- Live statistics: total students, pending ideas, active/total projects, total campaigns and total registrations.
- Tabular views of all registered students and all campaign registrations, joined with student and campaign details.
- An auto-generated Impact Report summarizing participation metrics, with a print/download option.

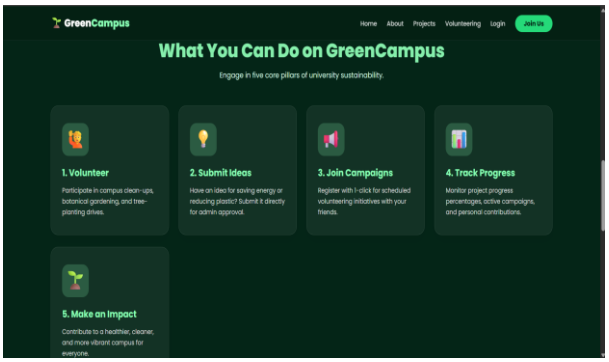
7. Screenshots

The following pages should be captured as screenshots and inserted below, showing both the Admin and Student/User interfaces as required by the assignment guidelines.

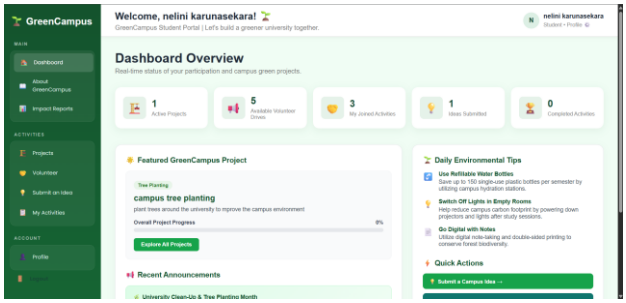
Login Page



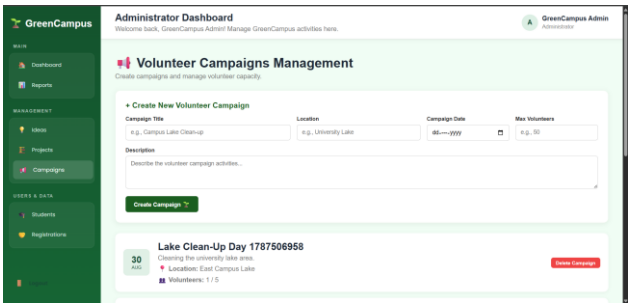
Student Registration Page



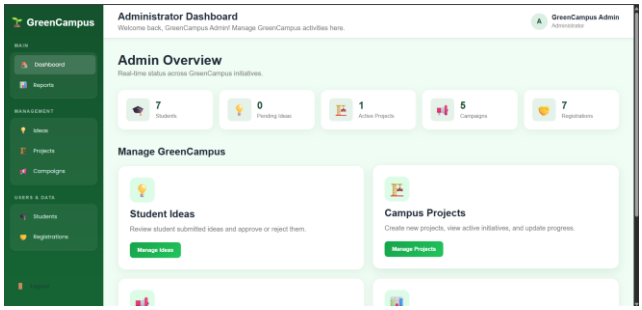
Admin Dashboard – Overview



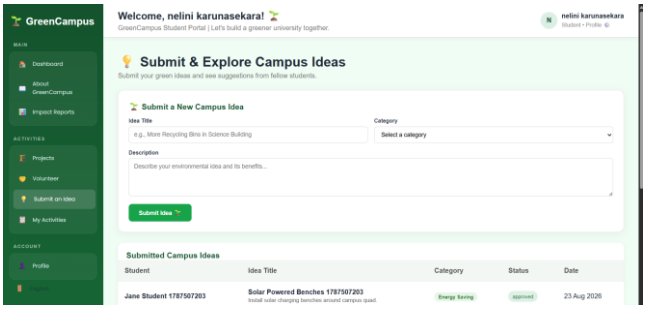
Admin – Manage Ideas (Approve/Reject)



Admin – Add Project



Admin – Add Campaign



Admin – Reports Section

Student Dashboard – Overview

Student – Submit an idea

Student/Public – Browse Project

Student – Browse & Join Campaigns

8. Source Code and Video Demonstration

Item	Link
Source Code (Google Drive, public access)	greennova.web.lk
Video Demonstration (Google Drive, public access)	https://nsbm365-my.sharepoint.com/:v:/g/personal/krasmeedum_students_nsbm_ac_lk/IQC-taz3xLiqSpufbVkRxIW1ATR-xpXOvrCoP2lWT6ToUrM?e=fl4KMc

9. Individual Contributions

Name	Contribution
K R A S Meedum	HTML, CSS, JavaScript + Flyer+ Report + Presentation + Web Hosting & Deployment
M P C J Jayasinghe	My SQL + Flyer + Presentation + Web Hosting & Deployment
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10. Challenges and Learning Outcomes

While developing our GreenCampus website, we faced several challenges throughout the project. At the beginning, it was difficult for us to decide exactly what features to include and how to connect all the parts of the system together.

One of the main difficulties we faced was **connecting the website with the database**. We had some problems with MySQL, table relationships, foreign keys, and database connections. We also faced issues when running our PHP files through XAMPP, especially with Apache, MySQL, ports, and file locations.

We also had some difficulties with the **login and registration system**. Making sure that user information was correctly stored and retrieved from the database required a lot of testing and debugging.

Another challenge was the **design of the website**. We wanted the website to look attractive and professional, so we had to make several changes to the layout, fonts, images, navigation bar, and dashboard before we were satisfied with the final result.

Since this was a **group project**, managing everyone's work was also challenging. We had to divide the tasks among our members, communicate with each other, and make sure that everyone's work could be combined without causing problems.

We also had to manage the project alongside our other university work, so **time management** was sometimes difficult. When errors occurred, we had to spend extra time finding the problem and trying different solutions.

Outcomes We Achieved as a Group

Even though we faced many difficulties, the project gave us a lot of valuable experience. We were able to successfully develop our **GreenCampus website** and bring our original idea into a working system.

We learned how to connect a website to a **MySQL database** and gained practical experience using **HTML, CSS, JavaScript, PHP, and XAMPP**. We also learned how to create features such as user registration, login, dashboards, and database-driven pages.

One of the biggest outcomes for us was improving our **problem-solving skills**. Instead of giving up when something did not work, we learned to identify the error, search for possible solutions, test them, and fix the problem.

The project also improved our **teamwork and communication skills**. We learned how important it is to share ideas, help each other when someone gets stuck, and coordinate our work properly.

Overall, this project gave us a better understanding of what it is actually like to develop a website as a team. **It was not always easy, but the problems we faced helped us learn more and become more confident in working with web development technologies.**

11. Conclusion

Green Campus successfully demonstrates a complete full-stack web application built with HTML, CSS, JavaScript, PHP and MySQL. The system fulfils the core requirements of the assignment: separate Admin and Student interfaces, secure authentication, full CRUD functionality across projects, campaigns and ideas, relational database integration, and a responsive, user-friendly design. Beyond the minimum requirements, the project adds real-time volunteer capacity tracking, an idea moderation workflow, and an auto-generated impact report, reflecting the goal of encouraging genuine student participation in campus sustainability initiatives.

Thank You!